



Event Coreference Resolution with Multi-Pass Sieves

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Event Coreference Resolution

- Determine which event mentions in a text refer to the same real-world event

Event Coreference Resolution

- Determine which **event** mentions in a text refer to the same real-world **event**

Much less studied than entity coreference resolution

Event Coreference Corpora

- MUC
- ACE
- OntoNotes
- ECB, ECB+
- IC
- KBP

Event Coreference Corpora

- MUC
- ACE
- OntoNotes
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- IC
- KBP
- Corpora differ in terms of
 - Domain (event types)
 - Joint ventures (MUC-6) vs. terrorist attacks (IC)

Event Coreference Corpora

- MUC
- ACE
- OntoNotes
- ECB, ECB+
- IC
- KBP
- Corpora differ in terms of
 - Domain (event types)
 - Number of event types
 - ACE has more event types than MUC

Event Coreference Corpora

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- KBP
- Corpora differ in terms of
 - Domain (event types)
 - Number of event types
 - Completeness of annotations
 - ECB not fully annotated (Liu et al. 2014)

Event Coreference Corpora

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- ACE
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- IC
- KBP
- Corpora differ in terms of
 - Domain (event types)
 - Number of event types
 - Completeness of annotations
 - Definitions of events & event coref
 - Differences between ACE & KBP

Three Major Differences between ACE and KBP Event Coreference

- ACE: single-word triggers
- KBP: additionally allows multi-word triggers
 - Jane was **laid off** by XYZ Corp.

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Words that express the occurrence of an event

Three Major Differences between ACE and KBP Event Coreference

- ACE: single-word triggers
- KBP: additionally allows multi-word triggers
- ACE: each word can trigger at most one event mention
- KBP: each word can trigger multiple mentions
 - “murder” can trigger two mentions
 - One with type **Life** and subtype **Die**
 - The other with type **Conflict** and subtype **Attack**

Three Major Differences between ACE and KBP Event Coreference

- ACE: single-word triggers
- KBP: additionally allows multi-word triggers
- ACE: each word can trigger at most one event mention
- KBP: each word can trigger multiple mentions
- ACE: strict notion of event coreference
- KBP: relaxed coreference criterion
 - Allows two event mentions to be coreferent as long as they **intuitively** refer to the same real-world event
 - E.g., “**attack in Baghdad on Thursday**” and “**bombing in the Green Zone last week**” can be coreferent in KBP but not ACE

Plan for the Talk

- KBP 2015 corpus
- Baseline system
- Our multi-pass sieve approach
- Evaluation

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KBP 2015 Corpus

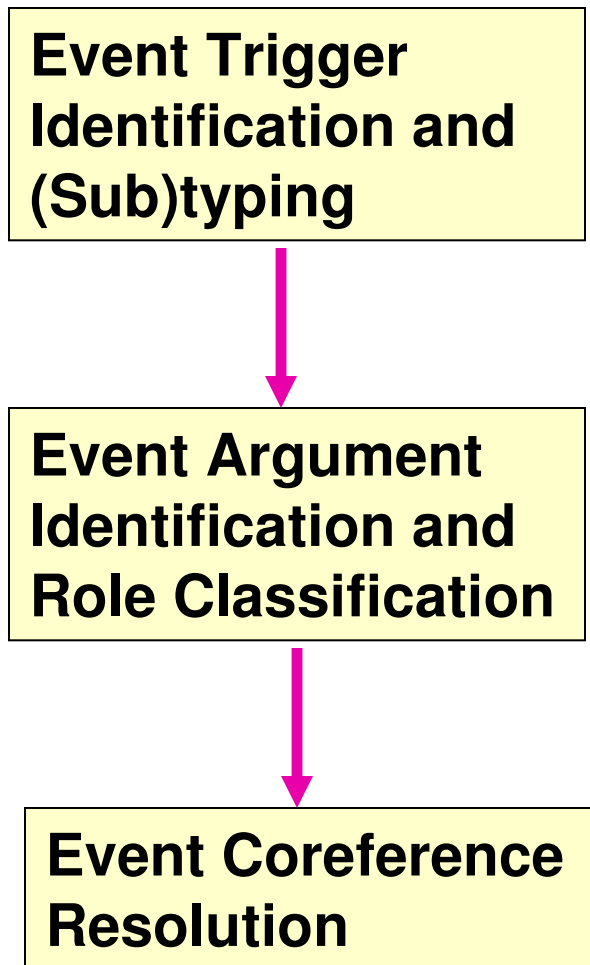
- more relaxed coreference criterion than ACE
- Each event has one of 33 subtypes
- Two types of documents
 - Newswire documents, discussion form documents

	Training		Test	
	Newswire	Forum	Newswire	Forum
Documents	81	77	98	104
Event mentions	2,219	4,319	3,788	2,650
Coref chains	1,461	1,874	2,440	1,685

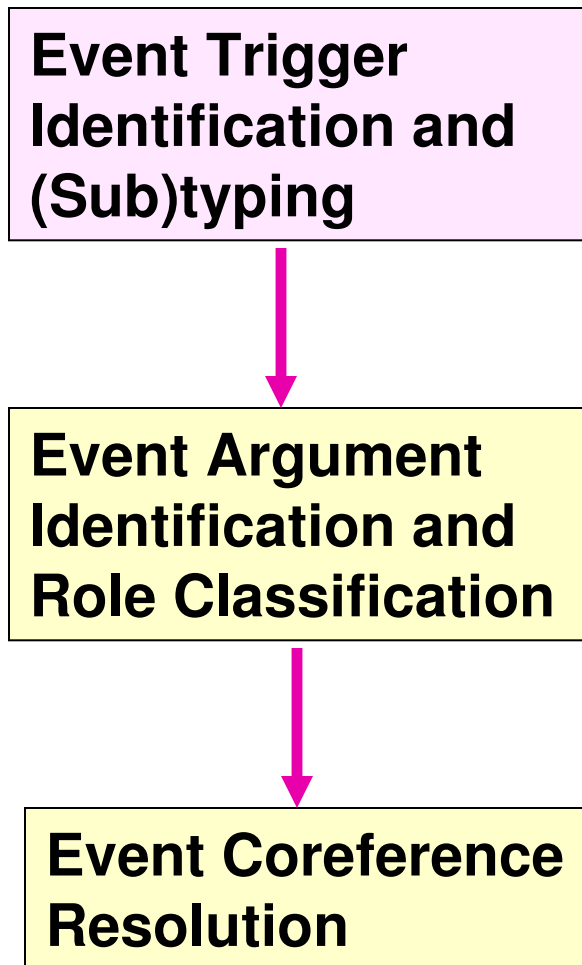
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Baseline System: Architecture

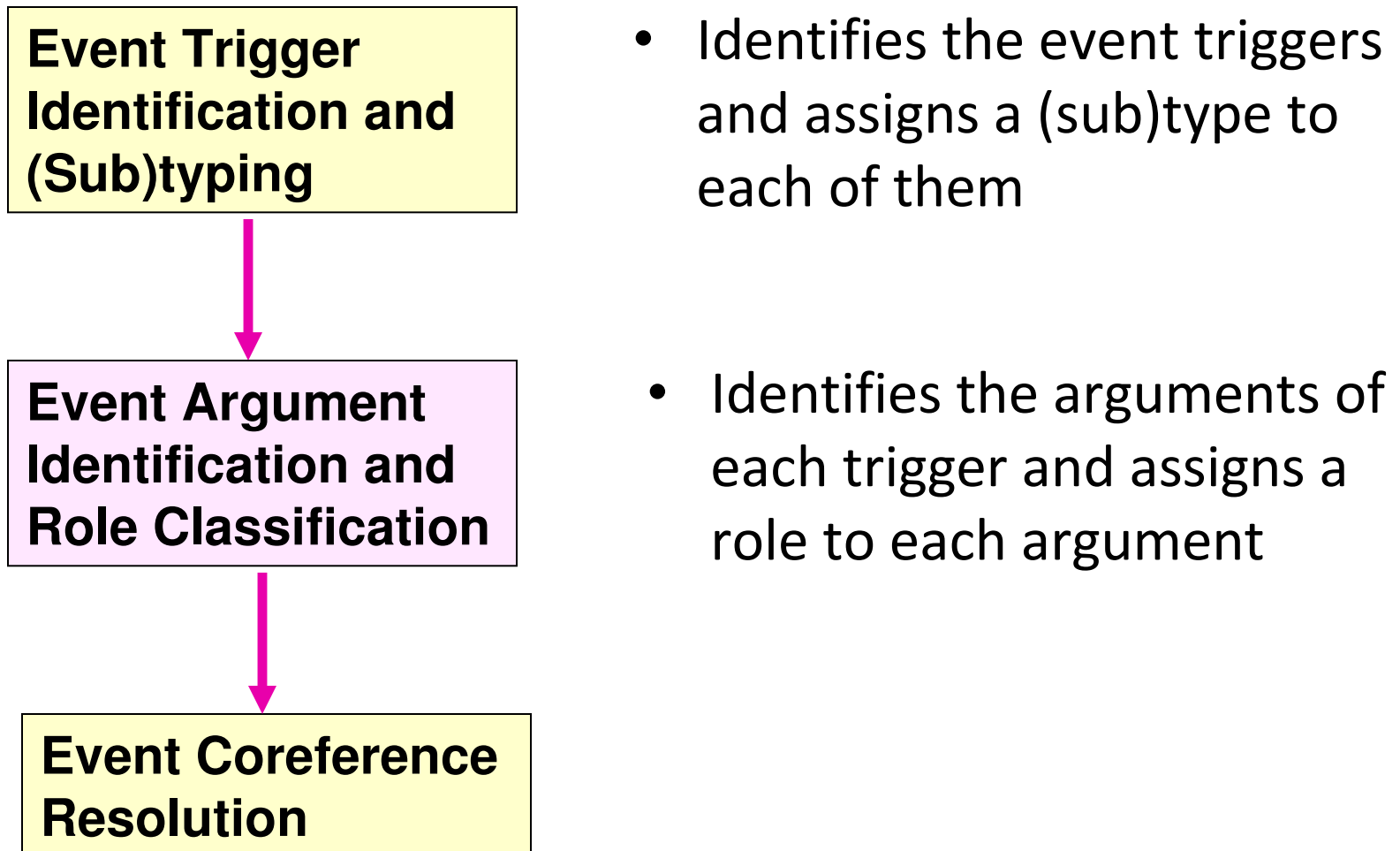


Baseline System: Architecture

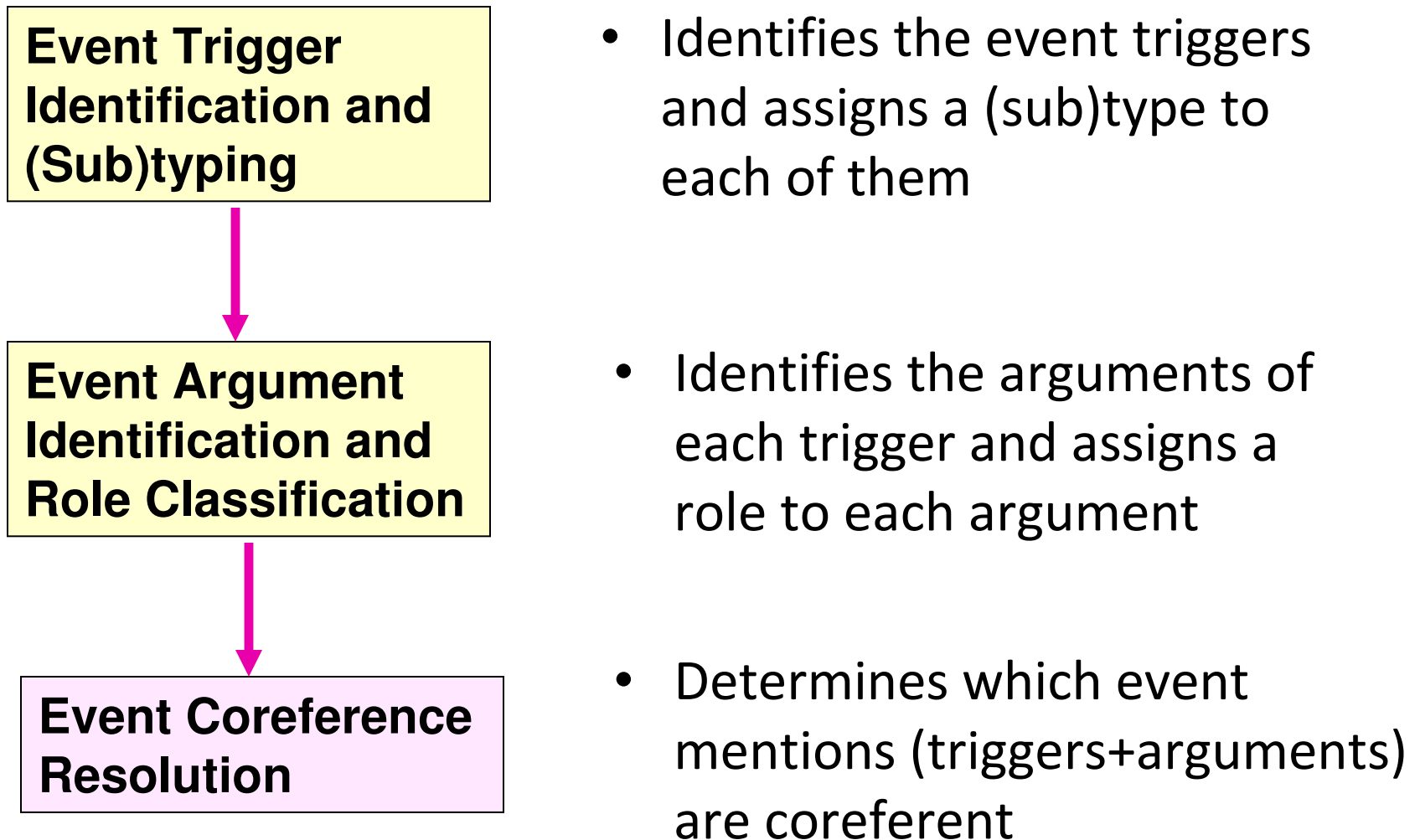


- Identifies the event triggers and assigns a (sub)type to each of them

Baseline System: Architecture



Baseline System: Architecture



Event Trigger Identification and (Sub)Typing

- Identifies the triggers and assigns a subtype to each of them
- Train a CRF for joint trigger identification & subtyping
 - Each instance corresponds to a word with 13 features
 - Class value: B-subtype, I-subtype, O
 - Train a CRF for each type
 - A trigger can have more than one type

Argument Identification & Role Classification

- Identifies the arguments for each trigger and assigns a role to each trigger
- Extracts the candidate arguments (i.e., entity mentions) in each document using a CRF
- Jointly learns argument identification & role classification using an SVM

Argument Identification & Role Classification

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- Extracts the candidate arguments (i.e., entity mentions) in each document using a CRF
- Jointly learns argument identification & role classification using an SVM
 - each instance is created by pairing a trigger with each of its candidate arguments and is represented using 9 features
 - Class value is the argument's role if the candidate argument is the trigger's true argument and null otherwise

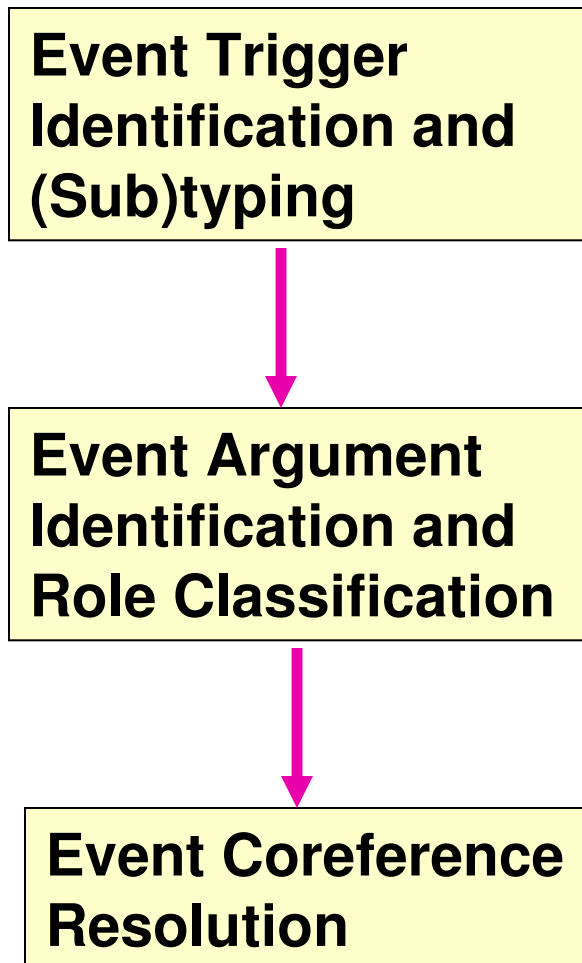
Event Coreference Resolution

- Learning-based
- Train a pairwise SVM classifier to determine whether two event mentions are coreferent
- During testing, cluster the event mentions using closest-first single-link clustering
 - Links each event mention to its closest preceding event mention

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Multi-Pass Sieve Approach



- First two components are the same as those of the baseline
- Differs from the baseline in terms of how the event coreference component was implemented

Multi-Pass Sieves: Overview

- Originally proposed for entity coreference resolution
- A sieve-based system is composed of a pipeline of sieves



- Each sieve is a (hand-crafted or learned) rule specifying the conditions for two mentions to be coreferent
 - E.g., two mentions are coreferent if they have the same head
- Sieves are arranged in **decreasing order of precision**
 - Coreference decisions made by earlier sieves are propagated to and can be exploited by later sieves

Multi-Pass Sieves for Event Coreference Resolution

- Challenge
 - Hand-crafting or learning high-precision rules for event coreference resolution is challenging

Our Approach: Six Sieves

- Tried to avoid designing different sieves for the two types of documents (Newswire and Discussion forum)
 - Only the first sieve is different for the two types of docs

Sieves for the Newswire Documents

- Sieve 1 (Headline sieve)
 - **Observation:** first sentence is almost always a detailed explanation of the headline
 - Posits an event mention in the headline and an event mention in first sentence as coreferent if:
 - they have the same subtype
 - their triggers are in the same WordNet synset

Sieves for the Newswire Documents

- Sieve 2 (Strict Event Coreference Sieve)
 - Posits two event mentions as coreferent if:
 - They have the same subtype
 - Their triggers have the same lemmatized form
 - There is a role for which the arguments of the two event mentions are in the same entity coreference chain or are lexically identical
 - Their triggers are in the same entity coreference chain if they are nouns

Sieves for the Newswire Documents

- Sieve 3 (Strict Trigger Match sieve)
 - Posits two event mentions with noun triggers as coreferent if:
 - They have the same subtype
 - Their triggers have the same lemma and same modifiers

Sieves for the Newswire Documents

- Sieve 4 (Relaxed Trigger Match sieve)
 - Relaxes the conditions in Sieve 2
 - Posits two event mentions as coreferent if:
 - They have the same subtype
 - There is a role for which the arguments of the two event mentions are in the same entity coreference chain or are lexically identical
 - Their triggers or their hypernyms are in the same WordNet synset

Sieves for the Newswire Documents

- Sieve 5 (Known Coreferent Pairs Sieve)
 - Memorizes the triggers that are coreferent in the training corpus
 - Posits two event mentions as coreferent if:
 - They have the same subtype
 - Their triggers have appeared in the training data as a coreferent pair

Sieves for the Newswire Documents

- Sieve 6 (Machine Learning sieve)
 - Posits two event mentions as coreferent if their coreference probability according to the baseline event coreference classifier exceeds a threshold

Sieves for the Discussion Forum Documents

- Same as the sieves for the newswire documents except for Sieve 1
- Sieve 1 for discussion forum documents
 - **Observation:** an author of a post often quotes a preceding post to which she wants to respond
 - Posits two event mentions as coreferent if the sentences containing their triggers are identical

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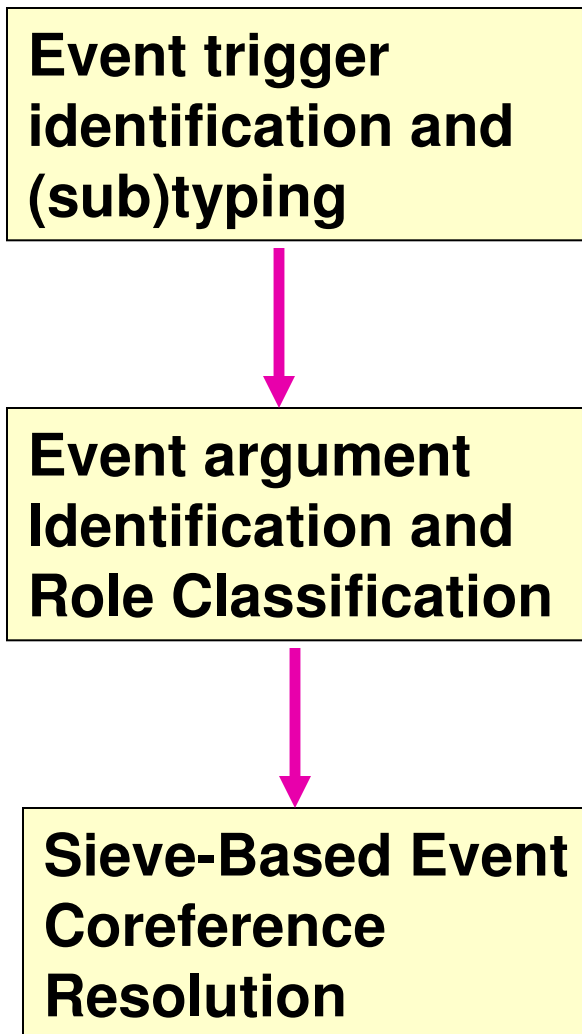
Experimental Setup

- End-to-end evaluation setting
 - Evaluated on automatically extracted event mentions

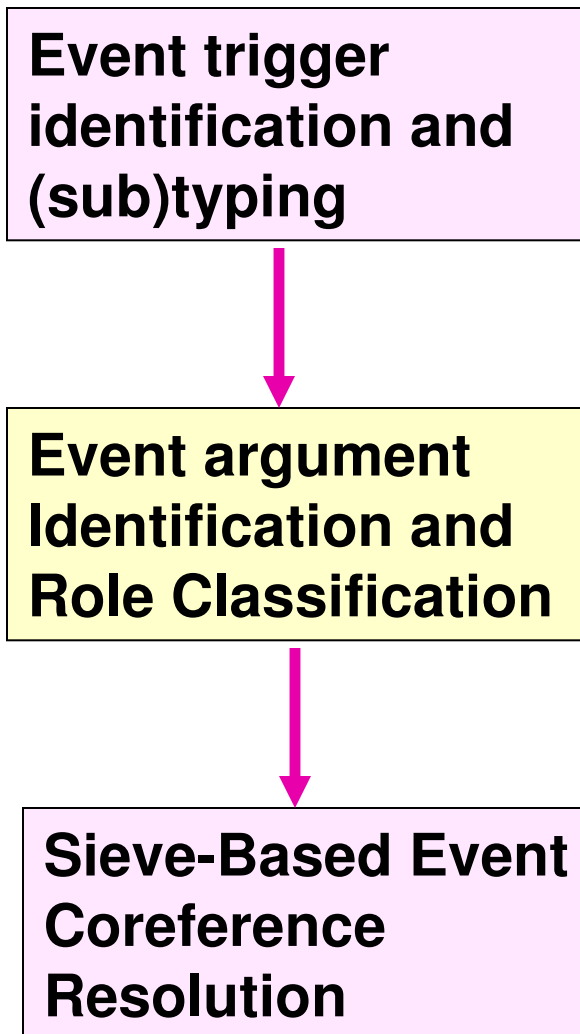
Experimental Setup

- End-to-end evaluation setting
 - Evaluated on automatically extracted event mentions
- Test set: official KBP 2015 evaluation data

Did we use the KBP training corpus?



Did we use the KBP training corpus?



KBP 2015 training data

Did we use the KBP training corpus?

**Event trigger
identification and
(sub)typing**

KBP 2015 training data

**Event argument
Identification and
Role Classification**

Two other LDC corpora

**Sieve-Based Event
Coreference
Resolution**

Official Evaluation Metric

- Unweighted average of
 - MUC
 - B^3
 - Entity-based CEAF
 - BLANC

Results: Baseline

	MUC	B ³	CEAF	BLANC	Avg
Baseline	37.1	45.5	38.5	30.1	37.8

Results: Sieve-based Approach

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Sieve 1	1.45	41.9	39.9	17.1	25.1

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Sieve 1	1.45	41.9	39.9	17.1	25.1
+ Sieve 2	11.2	42.8	40.8	19.9	28.7

- Sieve 2: Strict event coreference sieve

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+ Sieve 3	15.6	43.2	41.4	21.3	30.4

- Sieve 3: Strict trigger match sieve

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+ Sieve 4	18.3	43.4	41.3	22.2	31.3

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- Sieve 5: Known coreferent pairs sieve

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+ Sieve 5	43.9	45.4	38.7	33.1	40.3
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- Each sieve contributes positively to Avg F-score

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- Improvement of 2.5% F-score over Baseline

Comparison with KBP 2015 systems

- Our approach outperformed the best KBP 2015 participating system on the official test data
 - Ours: 40.32
 - Best KBP 2015 system: 39.65

Conclusions

- First sieve-based approach to event coreference
- Using a relatively simple set of sieves, we achieved the best result reported on the KBP 2015 test set
 - **Good news**: don't need sophisticated modeling
 - **Bad news**: low-hanging fruits, still fails on the hard cases
- Machine learning sieve contributed little to performance

Conclusions (Cont')

- Not the problem with machine learning
 - Too little training data for machine learning to be effective
 - Hard to learn when the event mentions of a given subtype appear only a few times in the training corpus
 - Word embeddings over-generalized
 - Training and test set distributions over event subtypes are different